

Single-Pilot Human Factors & Threat and Error Management



Most accidents are linked *to deficiencies in human performance.*

World wide statistics indicate that about 75% of aircraft accidents are caused by Human Factors (HF) deficiencies.



Objectives

The objective of this presentation is to provide guidance to instructors about teaching a realistic level of single pilot HF and TEM.

“Although these two subjects have a theoretical knowledge component, this presentation will concentrate on the application of these skills to flying”.



Overview

I. Single-pilot Human Factors

1. Maintain Effective Lookout
2. Assessing Effective Lookout
3. Maintain Situational Awareness
4. Assessing Situational Awareness
5. Assess Situations and Make Decisions
6. Assessing Decision-making
7. Set Priorities and Manage Task
8. Maintain Effective Communication and Interpersonal Relationships



II. Threat and Error Management

1. Threat And Error
2. SRM/CRM
3. Undesired Aircraft State
4. TEM application



Single-Pilot Human Factors



SINGLE PILOT HUMAN FACTORS



Achieving the safest outcome to flight operations by the most effective use of people and what people do when operating in the aviation environment and the equipment they use

Optimizing safe flight operations by enhancing the relationships between people, activities, and equipment.



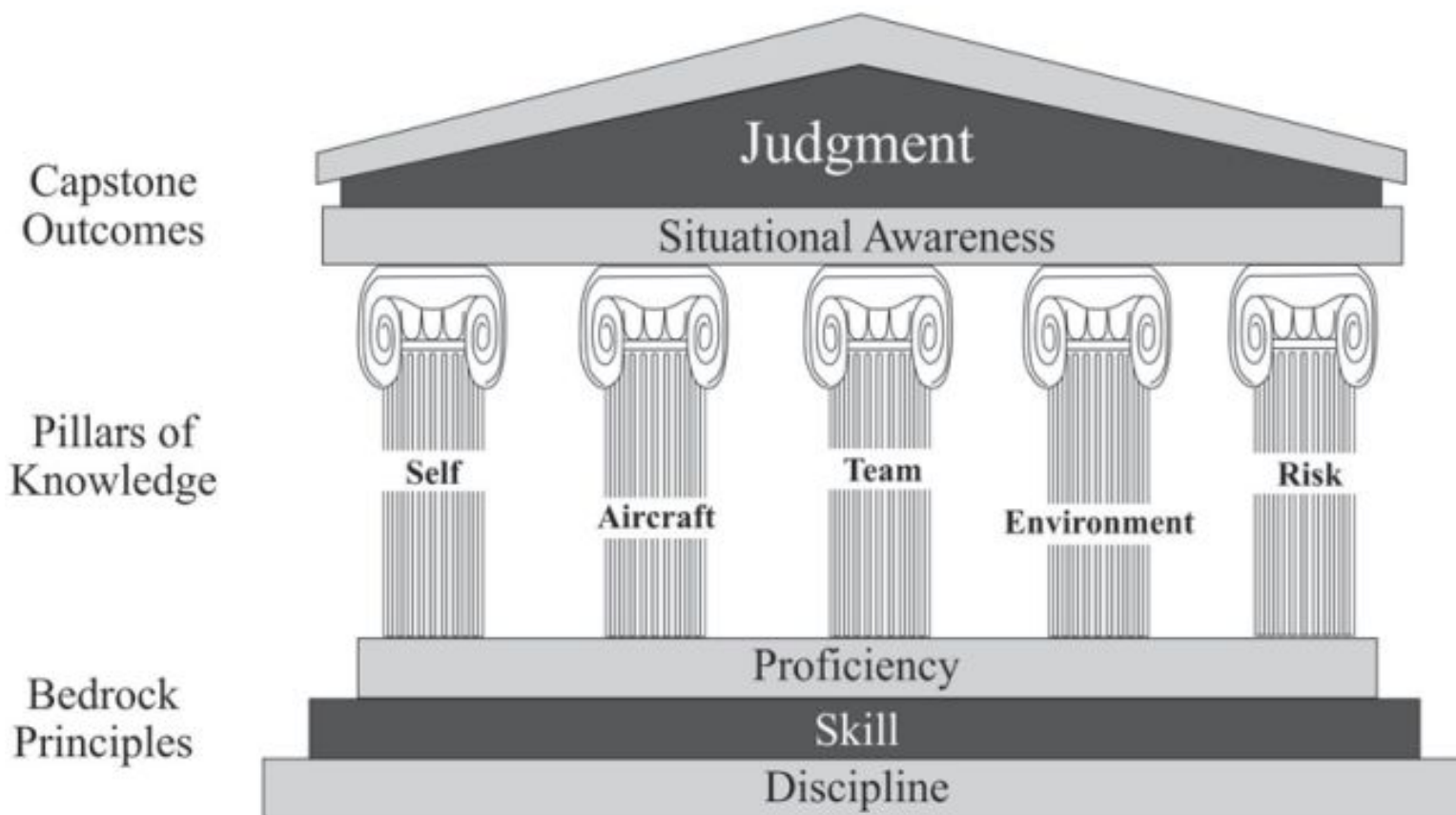
Airmanship



Airmanship is the consistent use of good judgment and well-developed skills to accomplish flight objectives. This consistency is founded on a cornerstone of uncompromising flight discipline and is developed through systematic skill acquisition and proficiency.

Airmanship covers a broad range of desirable behaviors and abilities in an aviator. It is not simply a measure of skill or technique, but also a measure of a pilot's awareness of the aircraft, the environment in which it operates, and of their own capabilities.





Pilots are required to continually process information during flight operations.

Stimuli are collected by the sensors: eyes, ears, nose, taste buds, skin and muscles (feel) and the vestibular senses (balance mechanism), and then this information is passed to the brain.

Some factors that may limit the construction of an accurate mental model are:

Experience: lack of experience will lead to the likelihood of not recognizing a stimulus.

Stress: may lead to single task fixation.

Anomalous perception: illusions, false signals from other people.

Lack of knowledge: can lead to a false premise



Information Processing

A decision is arrived at after the brain determines what to do about the options.

The *working* or *short term memory* holds the information being used at the time and may call on the *long term memory* to evaluate new information.

The brain is a single channel processor; If decisions are not prioritized correctly (the most critical decision first), the outcome could be unfavorable.



Limitations

Limitations that affect information processing and decision making are:

- Time limitations
- Mental overload, task mismanagement
- Conflicting information
- Expectations and anticipations
- Fatigue, forgetting, personality traits, stress
- Insufficient knowledge
- Emotions; confirmation bias (ignoring information that does not support the decision)
- Fixation and destination obsession



Single pilot human factors is composed of five (5) elements;

1. Maintain effective look out
2. Maintain situational awareness (SA)
3. Assess situations and make decisions
4. Set priorities and manage tasks
5. Maintain effective communications and interpersonal relationships



Threats are external to the aircraft

- The pilot must look outside the aircraft;
- Search the available visual field to detect threats that will probably appear in the peripheral vision;
- Shift vision directly to the threat and if identified as a collision risk, decide on what effective evasive action to take; and
- Maneuver the aircraft to mitigate the risk.



Assessing effective lookout

Assessment is the process of weighing evidence of an individuals performance against a standard

The evidence used must follow an established set of rules. These are:

Validity: It must cover all the performance criteria for the skills and knowledge of the standard being assessed.

Authenticity: It must be the individuals own work.

Sufficiency: Enough evidence must be collected to judge the individual is competent across all elements and performance criteria, all dimensions of competency.

Currency: The individual is competent now and meets the current standards.



Assessing Lookout

Lookout is a critical facet of safe flight operations and assessment of this skill will be ongoing through out a pilots career.

There are two main elements in effective lookout:

Firstly: to see an object

Secondly: to react appropriately to what has been seen.



Assessing Lookout

The three performance criteria relevant to maintaining an effective lookout are:

1. Maintains lookout and traffic separation using a systematic scan technique at a rate determined by traffic density, visibility, and terrain.
2. Maintains radio listening watch and interpret transmissions to determine traffic location and intentions; and
3. Performs “airspace cleared” procedure before commencing any maneuvers.

These three criteria must be achieved for a positive assessment of effective lookout.

These can be achieved by monitoring head and eye movements when possible and questioning the trainee on what they see.



MAINTAIN SITUATION AWARENESS (SA)

Situational Awareness is *The perception of the elements in the environment within a volume of time and space, the comprehension of their meaning and the projection of the status in the near future*

Three levels to maintain SA:

Level 1: Perception of the current environment

Level 2: Interpretation of the immediate situation

Level 3: Anticipation of the future environment

Monitoring and gathering information from both within the cockpit and outside the aircraft achieves perception of the current environment.



MAINTAIN SITUATION AWARENESS

Teaching Situation Awareness

Continued monitoring and verbalizing observations assists perception of what is happening and what is likely to happen in the near future, which is the basis of SA, visual information is the greatest source for building and maintaining SA.

How to re-establish SA whenever it is lost or degraded?

Observe and continue monitoring radio frequency; Contact ATC or other aircraft. “Questioning” are the primary means of making a formative assessment of SA.



Assessment in general

“Assessment tools” are the resources used by an assessor to gather evidence that a person is competent.

Examples of assessment tools are;

- Pilots logbook

- Examination results

- Training and achievement records

- Evidence and observation checklist

- Specific questions or activities

- Simulation and scenarios



These tools must be;

Valid: Assess what you came to assess using an approved standard in a realistic environment;

Reliable: A qualified assessor, consistent evidence gained from observation, questioning, simulation and training records, using clearly stated criteria and instructions;

Flexible; Assessment conducted in an operational environment using an aircraft in realistic flight circumstances with adjustments for different situations; and

Fair: Candidate's needs are identified and accommodated, any allowable adjustments catered for and appeal procedure explained.



ASSESSING SITUATION AWARENESS

The most important aspect of assessing Situational Awareness is to confirm that the pilots perception of the environment is accurate.

Next, find out what options have been generated and whether they are realistic.

Must determine if Situational Awareness is being maintained regardless of workload.

Must also observe the appropriate application or otherwise of knowledge, because Situational Awareness can be adversely affected by a lack of knowledge.

If Situational Awareness is observed below the required standard, record this deficiency as evidence for assessment.



The information process

- Receive information
- Convert information into reality
- Options are generated
- Options are analyzed
- A decision is made



Assessing decision - making

A good starting point for Assessors

- The pilot has to recognize that a decision has to be made and this procedure will require the pilot to gather and process information.
- The pilots action must be observable, but some questioning may be required to obtain an accurate assessment.
- Assessors must ensure the decision is the optimal one and is implemented effectively in the time available.
- It is acceptable to make a decision on the basis that it may require revision, if the safety of the flight is not compromised and continuously re-evaluate and update that initial decision.
- An assessor must observe a pilot to determine if he is able to manage the factors that limit perception which can adversely affect information processing and decision making.



SET PRIORITIES AND MANAGE TASKS

“aviate, navigate and communicate” is probably the basis of prioritization and task management.

Task management means completing a job or operation competently in the time available.

The brain is a single channel processor and humans can normally only manage one activity at a time.

During flight, assessor must be encouraged to prioritize tasks to ensure that the important and safety critical actions are dealt with first.

Managing stress is good organization in the cockpit and achieving an appropriate work rate.



ASSESSING PRIORITISATION AND TASK MANAGEMENT

- An assessor must be able to assemble evidence of competence in setting priorities and managing tasks on a flight test by simply observing a pilot's work pattern and task completion. The danger is that such assessment is prone to subjectivity. Valid evidence must be obtained.
- When assessing task management the assessor must be looking for competent completion of a task in the time available.
- The assessment process will require detailed observation, information gathering and questioning because there will be a need to determine how a candidate's mind is functioning while managing tasks. By obtaining this information and combining it with observations it is possible to judge a pilot's ability to competently set priorities and manage tasks.



Factors that adversely affect a pilot's ability to operate efficiently

- Lack of preparation (confusion, disorganization)
- Fatigue (poor decision making, errors)
- Discomfort (distraction, fatigue)
- Stress (inefficiency, distraction)
- Arousal (increased or decreased work cycles)
- Domestic stress (distraction, lack of concentration)
- Distraction (diverted attention)
- Destination or task obsession (poor decision making)
- Bad health (decreased physical and psychological performance)
- Over load (fixation, tunnel vision, broken work cycles)



Maintain effective communication and interpersonal relationships

The intent of the “*maintain effective interpersonal relationships*” component of the element is to make pilots aware of the need to always foster positive and cooperative relationships with persons involved with or affected by the flying operations to be performed.

Communication is not limited to the radio telephone; it also involves direct verbal and non-verbal exchanges.

Effective interpersonal relationships involves being able to get a positive or helpful rather than negative or obstructive response from individuals or groups that a pilot deals with. A major component of interpersonal relationships is effective communication.

The first requirement for communication is a common language and “*Aviation Phraseology*”



Maintain effective communication and interpersonal relationships

Some personal characteristics that must be evaluated are:

- Tone and phrasing of communication
- Openness
- Reaction to criticism
- Aggressiveness or lack of assertion
- Willingness to listen
- Respect for others; arrogance, and use of authority



Performance criteria:

- The first performance criteria for the element is: “Establishes and maintains effective and efficient communications and interpersonal relationships with all stake holders to ensure a safe outcome of the flight”.
- The second performance criteria for the element is: “Defines and explains objectives to applicable/involved stakeholders.”
- The third performance criteria for the element is: “Demonstrate a level of assertiveness that ensures the safe completion of the flight.”
- The fourth performance criteria for the element is: “Encourages passengers to participate in and contribute to the safe outcome of the flight.”



Threat And Error Management



What is Threat and Error Management?

Threat and Error Management (TEM) is a risk management or risk mitigation. It is a process of detecting and responding to threats and errors to prevent safety being compromised.

Left unmanaged, these threats and errors can lead to '**undesired aircraft states**'- the last opportunity to avoid a serious incident or accident. TEM can be considered an extension to airmanship, providing a structured way to help people maintain safety margins during everyday operations.

The safety benefit of recognizing and managing threats, errors and undesired states before they lead to a serious incident or accident is of course not just limited to flight crews. TEM is equally applicable and relevant to cabin crew, air traffic control (ATC) and maintenance operations.



The Goals of Threat and Error Management

- Identify & Recognize Threats
- Reduce Errors
- Prevent Undesired A/C States (UAS)
- Help recognize and prevent the accumulation of mistakes which can escalate into operational errors
- Break the chain early (Accidents are rarely the result of a single event or error)



Why is Threat and Error Management important?

Without TEM, what is the worst that can happen? TEM complements risk management processes within safety management systems and can assist in maintaining the safety of the operation and ultimately keep people safe.

There are numerous examples of fatal accidents that have occurred due to decision-making errors, such as continuing a visual flight into deteriorating weather conditions, which could have been prevented by thorough preflight planning

Air accidents have also occurred due to errors made in the maintenance hangar – leading to catastrophic consequences in flight – which could have been identified before the aircraft ever left the ground

After an accident it is, of course, important to understand why the errors occurred and how the threats went undetected, but what is equally as important is preventing these threats and errors from developing into adverse outcomes, such as described above.



Components of Threat & Error Management

1. Threat
2. Error
3. SRM
4. Undesired A/C state



What is a Threat?

An outside (**uncontrollable**) event that increases the operational complexity of a flight

Events errors that occur beyond the influence of the flight crew, increase operational complexity and which must be managed to maintain the margin of safety.

Threats can be defined as a situation or event that has the potential to have a negative effect on safety. Threats can be classified as **internal and external**.



What is a Threat?

Threats can also be categorized as **anticipated, unanticipated, and latent threats**, and all three can reduce safety margins.

Anticipated threats are those that can be predicted, such as thunderstorms, congested airports, and complex or physically demanding tasks.

Unanticipated threats are those that occur unexpectedly, such as unforeseen turbulence or unexpected equipment malfunctions.

Latent threats, however, are those that may not be directly obvious or observable and may need to be discovered through formal safety analysis. These may include socio psychology and/or organizational factors such as organizational culture, operational pressures and normalized behaviors.



Internal Threats

Internal threats are related to the individual.

Examples of internal threats are:

- Fatigue
- Experience
- Attitude
- Lack of recency and proficiency
- Health and wellbeing.
- Hazardous behavior
- Over or under confidence
- Lack of flight discipline



External Threats

External threats can relate to the context of the operation and therefore can be different depending on the situation.

Examples of external threats within different operational context are:

Pilot-related threats-

- adverse weather
- high terrain or obstacles (wire, etc.)
- night operations
- other traffic
- equipment faults
- remote strips/landing sites
- weight and balance.

-ATC-related threats-

- airspace design
- airport layout
- co-ordination
- communication failure
- navigation aids
- traffic
- visibility/weather.



External Threats

-Cabin crew-related threats-

- unanticipated turbulence
- unruly passengers
- smoke in cabin
- equipment faults
- Interruption/distraction
- lithium battery (PED, etc.) fire
- sudden depressurisation.

Maintenance-related threats-

- shift handover
- uncontrolled tooling
- environment, lighting, etc.
- night shifts
- poor documentation
- lack of tooling
- time pressure.



Other Threats:

- Operational Threats
 - Equipment malfunction, closures on an airfield
- Environmental Threats
 - Weather, ATC
- Mismanaged Threats
 - Inappropriate control inputs
 - Not extending the landing gear
 - Not handling an engine out (single or multi)
 - Improper speed for flight condition

Threats not Addressed

- Improperly addressed threats will often introduce another threat
- Properly addressed threats help mitigate the impact of the event of situation



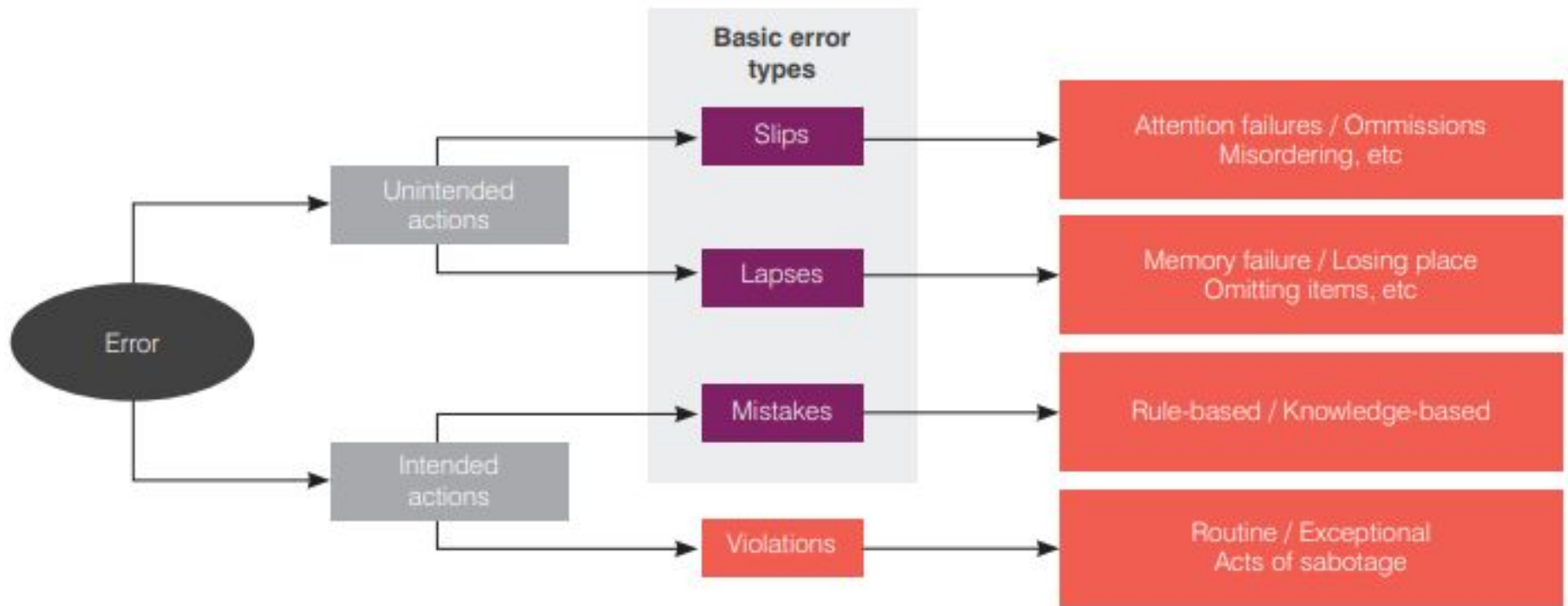
What is an Error?

As humans we are fallible, and errors are to be expected. Even the most experienced and well-trained person can make an error. Errors can arise from observable events or situations as well as from decisions made about those situations. Errors can be classified as slips, lapses, or mistakes, and are generally always considered unintentional. External and internal threats may lead to errors on the part of the pilot.

Errors can be defined as actions or inactions which can lead to:

- ❑ A deviation from individual or organisational intentions or expectations
- ❑ Reduces safety margins
- ❑ Increase the probability of adverse operational events on the ground or in flight
- ❑ Can arise from observable events or situations as well as from decisions made about those situations





Three Categories of Errors

Aircraft Handling Errors

- ❑ Speed, configuration, automation
- ❑ Incorrect systems operation or management

Procedural Errors

- ❑ Incorrect performance calculations
- ❑ Inaccurate flight planning
- ❑ Checklist error

Communication Errors

- ❑ Misunderstandings
- ❑ Non-standard communication
- ❑ Pilot and ATC
- ❑ Between Crew
- ❑ Between Crew and Passengers



Single Pilot Resource Management



Single-Pilot Resource Management

the basic idea of single-pilot resource management (SRM) and a subject treated as a subsidiary concept, aeronautical decision making (ADM) In 2020, both are considered essential to effective safety enhancements (SEs) in general aviation. Some aviation professionals colloquially refer to SRM as single-pilot CRM because of its historical connection to crew resource management used by airline flight crews.



Definitions

Single-pilot resource management (SRM) — The art and science of managing all the resources — both on-board the aircraft and from outside sources — available to a single pilot prior to and during flight, to ensure the successful outcome of the flight. structured approach to SRM helps pilots learn to gather information, analyze it, and make sound decisions on the conduct of the flight.

Aeronautical decision making (ADM) — A systematic approach to the mental process used by aircraft pilots to consistently determine the best course of action in response to a given set of circumstances.

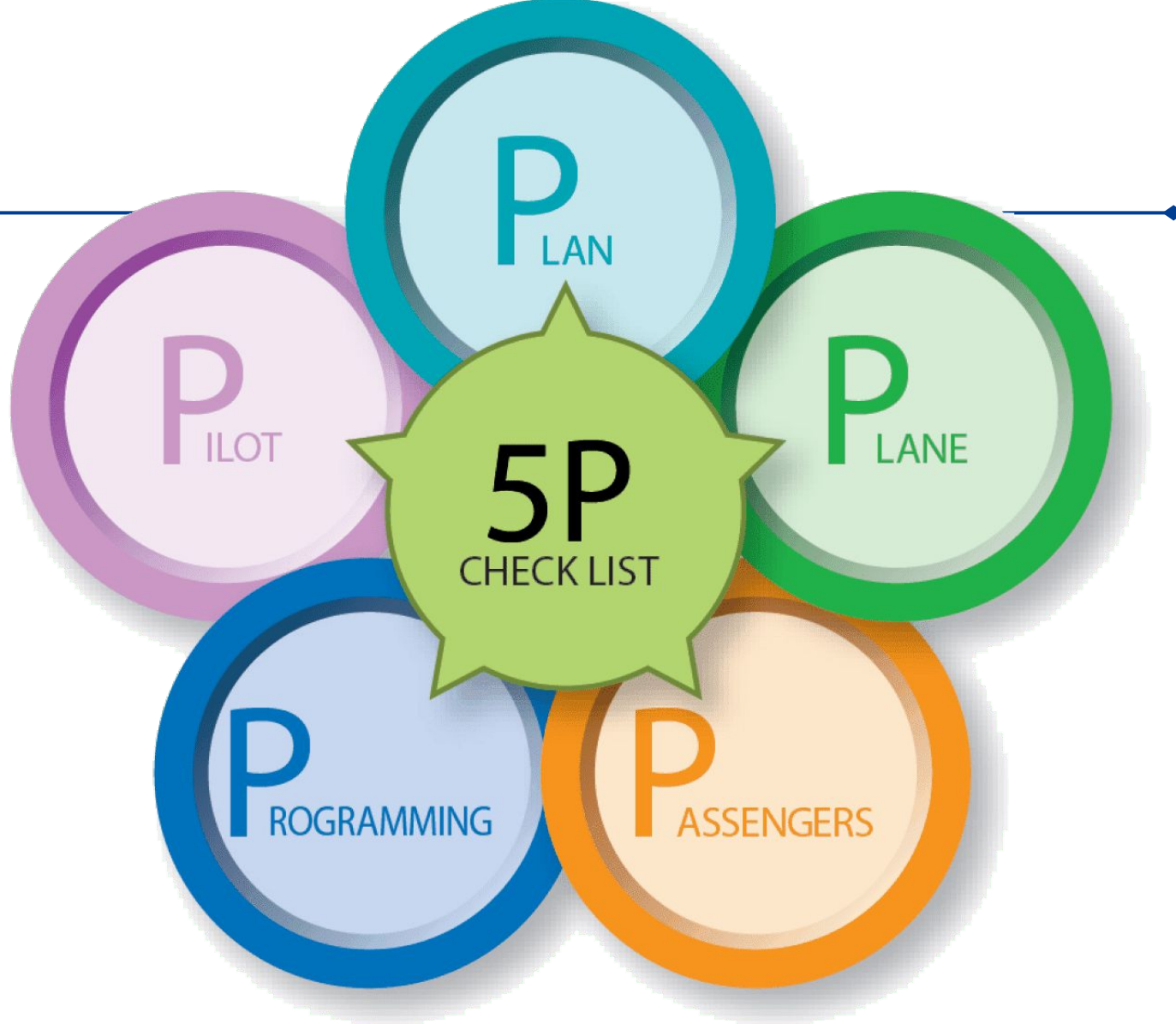


Single Pilot Crew Resource Management

There is no one right answer in aeronautical decision-making. Each pilot is expected to analyze each situation in light of experience level, personal minimums, and current physical and mental readiness level, and make his or her own decision.

To get the greatest benefit from SRM, you also need a practical framework for application in day-to-day flying. One such approach involves regular evaluation of the **5 P's**





5 P's Approach

Plan - The plan includes the basic elements of crosscountry planning: weather, route, fuel, current publications, etc. The plan also includes all the events that surround the flight and allow the pilot to accomplish the mission. The pilot should review and update the plan at regular intervals in the flight, bearing in mind that any of the factors in the original plan can change at any time.

Plane - The plane includes the airframe, systems, and equipment, including avionics. The pilot should be proficient in the use of all installed equipment as well as familiar with the aircraft/equipment's performance characteristics and limitations. As the flight proceeds, the pilot should monitor the aircraft's systems and instruments in order to detect any abnormal indications at the earliest opportunity.

Pilot - The pilot needs to pass the traditional "IMSAFE" checklist. This part of the 5P process helps a pilot identify and mitigate physiological hazards at all stages of the flight.



5 P's Approach

Passengers - The passengers can be a great help to the pilot by performing tasks such as those listed earlier. However, passenger needs — e.g., physiological discomfort, anxiety about the flight, or desire to reach the destination — can create potentially dangerous distractions. If the passenger is a pilot, it is also important to establish who is doing what. The 5P approach reminds the pilot-in-command to consider and account for these factors.

Programming - The programming can refer to both panelmounted and hand-held equipment. Today's electronic instrument displays, moving map navigators, and autopilots can reduce pilot workload and increase pilot situational awareness. However, the task of programming or operating both installed and handheld equipment (e.g., tablets) can create a serious distraction from other flight duties. This part of the 5P approach reminds the pilot to mitigate this risk by having a thorough understanding of the equipment long before takeoff, and by planning in advance when and where the programming for approaches, route changes, and airport information gathering should be accomplished, as well as times it should not be attempted.



Crew Resource Management

Decision making
CRM Acronyms Workload Management
Situational Awareness Speaking up
Briefing **Communication**
Leadership Error management
Red Flags Re-Evaluation
Closed Loop Communication Teamwork
Stressmanagement



Crew Resource Management

Crew Resource Management (CRM) is the effective use of all available resources for flight crew personnel to assure a safe and efficient operation, reducing error, avoiding stress and increasing efficiency.

Crew resource management or **cockpit resource management (CRM)** is a set of training procedures for use in environments where human error can have devastating effects. CRM is primarily used for improving aviation safety and focuses on **interpersonal communication**, **leadership**, and **decision making** in aircraft cockpits.



Crew Resource Management

CRM was developed as a response to new insights into the causes of aircraft accidents which followed from the introduction of flight data recorders (FDRs) and cockpit voice recorders (CVRs) into modern jet aircraft. Information gathered from these devices has suggested that many accidents do not result from a technical malfunction of the aircraft or its systems, nor from a failure of aircraft handling skills or a lack of technical knowledge on the part of the crew; it appears instead that they are caused by the inability of crews to respond appropriately to the situation in which they find themselves. For example, inadequate communications between crew members and other parties could lead to a loss of situational awareness, a breakdown in teamwork in the aircraft, and, ultimately, to a wrong decision or series of decisions which result in a serious incident or a fatal accident.



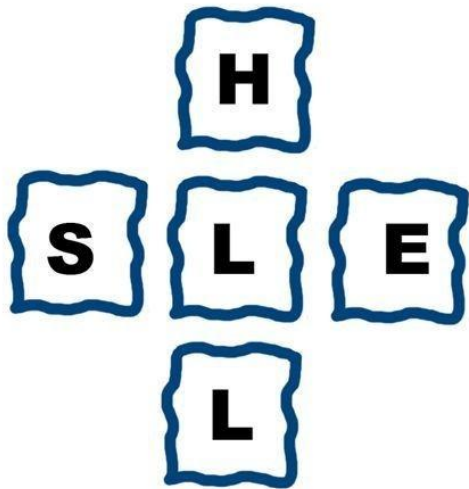
Crew Resource Management

- CRM encompasses a wide range of knowledge, skills and attitudes including communications, situational awareness, problem solving, decision making, and teamwork; together with all the attendant sub-disciplines which each of these areas entails. The elements which comprise CRM are not new but have been recognized in one form or another since aviation began, usually under more general headings such as 'Airmanship', 'Captaincy', 'Crew Co-operation'
- CRM can therefore be defined as a management system which makes optimum use of all available resources - equipment, procedures and people - to promote safety and enhance the efficiency of flight operations.
- CRM is concerned not so much with the technical knowledge and skills required to fly and operate an aircraft but rather with the cognitive and interpersonal skills needed to manage the flight within an organized aviation system. In this context, cognitive skills are defined as the mental processes used for gaining and maintaining situational awareness, for solving problems and for taking decisions. Interpersonal skills are regarded as communications and a range of behavioral activities associated with teamwork



SHEL MODEL

The SHEL Model



Interrelationship between human factors and the aviation environment

- ❖ **Software** - the rules, procedures, written documents etc., which are part of the standard operating procedures.
- ❖ **Hardware** - Machine, the Air Traffic Control suites, their configuration, controls and surfaces, displays and functional systems.
- ❖ **Environment** - the situation in which the L-H-S system must function, the social and economic climate as well as the natural environment.
- ❖ **Liveware** - the human beings - flight crews, ATC controllers, engineers and maintenance personnel, management and administration people - within in the system.



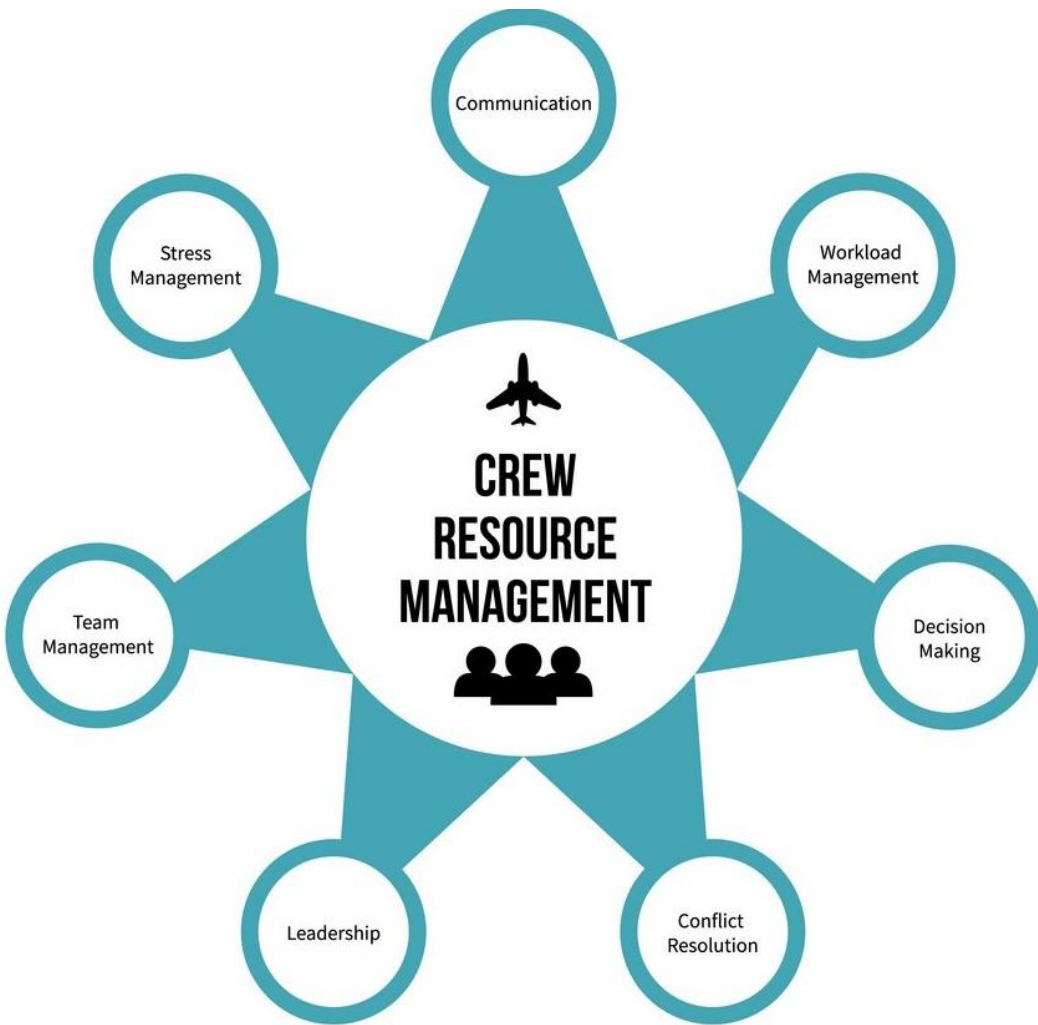
Liveware - In the centre of the model is the person or the human being. It is not a coincidence that the person is placed in the centre – humans are the most critical and most flexible component in the system. Remembering that in 75% of incidents and accidents, human error is the primary cause.

- ❑ **Liveware-Software (people and software – procedures, checklists, rules)** – In order to achieve a safe, effective operation between the liveware and software it is important to ensure that the software, particularly if it concerns rules and procedures, is capable of being implemented. That they are easy to understand, simple and easy to use.
- ❑ **Liveware-hardware (people and machine)** - This interface is the one most commonly considered when speaking of human-machine systems. displays that match the symbology that the user is familiar with. i.e. a warning light that normally is red has been fitted with a green cover. This could lead to confusion and failure to discover the warning if activated.



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- ❑ **Liveware – Environment (people and the environment)** - The liveware - environment interface refers to those interactions, which may be out of the direct control of humans, namely the physical environment - temperature, weather.
 - ❑ **Liveware-Liveware (the interface between people and other people)** - In this interface, we are concerned with leadership, cooperation, teamwork and personality interactions. Staff and management relationships are also included as a company culture and climate along with operational pressures has affects on human behaviour and performance.





Causes of Poor Management of Resources

- ❑ Lack of proper CRM training. In the absence of good training, crews may exhibit poor resource management and have an inadequate understanding of the value of CRM.
- ❑ Poor technical knowledge. Not having sound technical knowledge can affect one's performance adversely and also result in confusion, self-doubt, lack of confidence and credibility problems.
- ❑ The legacy of national culture. Some cultures may have very high authority gradients, making it difficult for a junior person to question the decisions and actions of a more senior person (e.g., a first officer being reluctant to speak up against a risky decision by the captain).
- ❑ Misunderstandings due to cultural, attitudinal or linguistic differences. Even when there is the intent to “get along,” these factors can lead to inadvertent resource management problems.



Causes of Poor Management of Resources

- ❑ Disruptive organizational culture. Some companies have a culture of discouraging people from speaking out. They may also place undue operational pressures on their flight crews that can lead to suboptimal decision making.
- ❑ Individual personality and attitude. Some people, by nature, do not want to listen to others and shun anyone who speaks up with an opinion contrary to theirs.
- ❑ Emotional status. A very negative experience, personal or occupational, can degrade resource management skills.
- ❑ Past experience. If someone experienced a bad result from speaking up and was chastised for it, that person may be less inclined to speak out again, even if he or she feels that the situation warrants it.
- ❑ Not appreciating the seriousness of the situation. Poor technical and/or human factors knowledge, as well as high workload and stress, can cause someone to misconstrue the gravity of events.



Causes of Poor Management of Resources

- Fatigue. Being tired can affect a person enough to turn a normally affable, cooperative person into someone who is very difficult to deal with. Fatigued individuals are also more prone to fixations, low situational awareness and poor decision making.
- Poor attitude. If someone has an inflated view of himself, his resource management will likely suffer. For example, believing one is almost invincible or infallible makes it difficult to work as a member of a team.
- A “them and us” attitude toward other aviation workers. Teamwork is built on mutual respect. Pilots certainly have a safety-critical job, but they cannot perform it effectively without help from the other team members.
- Prejudices against people. If someone has a negative attitude toward another person because of factors such as his or her background, culture or gender, it is less likely that interactions with that person will be positive.



How to Prevent Unsafe Conditions

- Good background knowledge and understanding of human factors and safety. Understanding how people work and interact is an important prerequisite for managing crew resources.
- A good CRM training course. Such a course should cover situational awareness, decision making, threat and error management, teamwork (including communication, cooperation, leadership and followership), human-machine interaction and automation.
- Sound English language skills. This is particularly important in a multicultural environment because English is the common language of aviation.
- Awareness of one's own national or organizational leanings. For instance, if you are working for a particularly hierarchical organization with a very high authority gradient where you are discouraged from speaking out to your seniors, you need to make a conscious effort to speak out when safety may be at stake.



How to Prevent Unsafe Conditions

- Awareness of one's own tendencies and attitudes. For instance, if you tend to take offense at having someone tell you that he or she does not agree with you, you may need to rethink your reactions and make particular efforts to be more accepting about listening to differing opinions.
- Proper planning and briefing, and not deviating from standard operating procedures (SOPs). This will reduce the chance of a misunderstanding.
- Improved technical knowledge and expertise. This will help you deal more confidently with many situations without becoming overloaded. It will also permit you to work on a problem with others in a less defensive manner if your opinion is questioned.
- Awareness of other people's cultures, backgrounds and attitudes. Awareness is a vital step in developing a better and more tolerant understanding of other people.



How to Prevent Unsafe Conditions

- ❑ Do not assume that other people have the same thought process, opinions or cultural leanings as you do. Everyone is different, and the way you do things or the way you think is not the only “right” way. Be assertive but, at the same time, remain open and recognize the potential for legitimate differences.
- ❑ Do not assume that other people have noticed a problem. If it is not absolutely clear that they have noticed it, point it out to them diplomatically and respectfully.
- ❑ Do not succumb to operational pressure if doing so may compromise safety. Be aware that operational pressure can influence your decisions, and be alert for any adverse effects from them. If in doubt, speak out.
- ❑ Do not be afraid to ask for help. Use all available resources, and do not make the problem worse by keeping it to yourself.



How to Prevent Unsafe Conditions

- Practice better decision making and communication in your daily life. Focus on maintaining maximum situational awareness. You do not need to be aviating to practice those things.
- If you identify shortcomings in your own behavior, have a trusted friend keep his or her eye on you and point out to you when you do something undesirable. It will help you to know what you are doing and how you are improving. It does not matter where you do this. It does not even have to be related to aviation.
- Do not be prejudiced against others. Be respectful toward others and embody the spirit of teamwork.
- Remember, the common goal of all aviation professionals is safety. You and everyone you work with must share the desire and objective of doing the job safely and contributing to better aviation safety records.



Crew Resource Management

CRM training is all about raising the issue of human error without pointing the finger of blame at any individuals.

Machines are logical - humans are not. We can be rude or unpleasant to a machine, and it will still provide the correct information provided we have pressed the right buttons and controls. But people are different, we are emotional individuals. Emotions are a fact as far as human are concerned. And it is most often our emotions that inhibit a free flow of information despite the common knowledge, that communication and sharing information is vital. Many accidents have showed, that individuals of a team were in possession of vital bits of information that never saw the light.

It will never just be the leader or the Captain who fails - it is the group as a whole. Because it is the group as a whole that holds all the relevant bits of information. However it is important that any leader (or trainer) develops an in-depth understanding of humans, our capacities and limitations as well as an increased understanding of communication enabling them to stimulate information sharing.



Aeronautical Decision Making



Aeronautical Decision Making (ADM)

ADM is a systematic approach to the mental process used by aircraft pilots to consistently determine the best course of action in response to a given set of circumstances

Decision making is the cognitive process of selecting a course of action from among multiple alternatives. The decision-making process produces a choice of action or an opinion that determines the decision maker's behavior and therefore has a profound influence on task performance.

Decision making in an aeronautical environment involves any pertinent decision a pilot must make during the conduct of a flight. It includes both preflight go/no-go decisions as well as those made during the flight. In aeronautics, decision making is of particular importance because of the safety consequences of poor decisions.



Aviation is a complex, safety-critical endeavor. Many decisions made while flying can affect the lives of hundreds of people and have extraordinary economic consequences. Thus, even though some flight decisions are not strongly related to safety, it is best to view ADM as a safety-critical function.

Decision making in aeronautics builds upon the foundation of conventional decision making. However, that ADM is carried out in dynamic and complex environments often characterized by:

- Ill-structured problems
- An abundance of information
- Uncertainty
- Shifting, ill-defined or competing goals
- Multiple event-feedback loops
- Time constraints
- High stakes with high levels of risk
- Collaboration and task sharing among multiple players
- Organizational norms and goals that must be balanced against the decision maker's personal choices



Decisions in such a complex environment should involve the following considerations:

- A decision is not unique but, instead, is a series of multiple and interdependent decisions that are made in real time and in a continuously changing autonomous environment.
- A human being is not able to perceive, evaluate, understand and act on all aspects of the environment. The decision maker must simplify reality and make a decision within it. This mechanism is called "bounded rationality."
- The principle of sufficiency describes a decision as a continuous process in which a set of decisions is made while seeking satisfactory results to a given situation. This principle does not mean the decision involves the least cognitive effort but, rather, that the human being has achieved a satisfying response to the situation.



The DECIDE Model

Detect that a change has occurred.

Estimate the need to counter or react.

Choose the desired outcome.

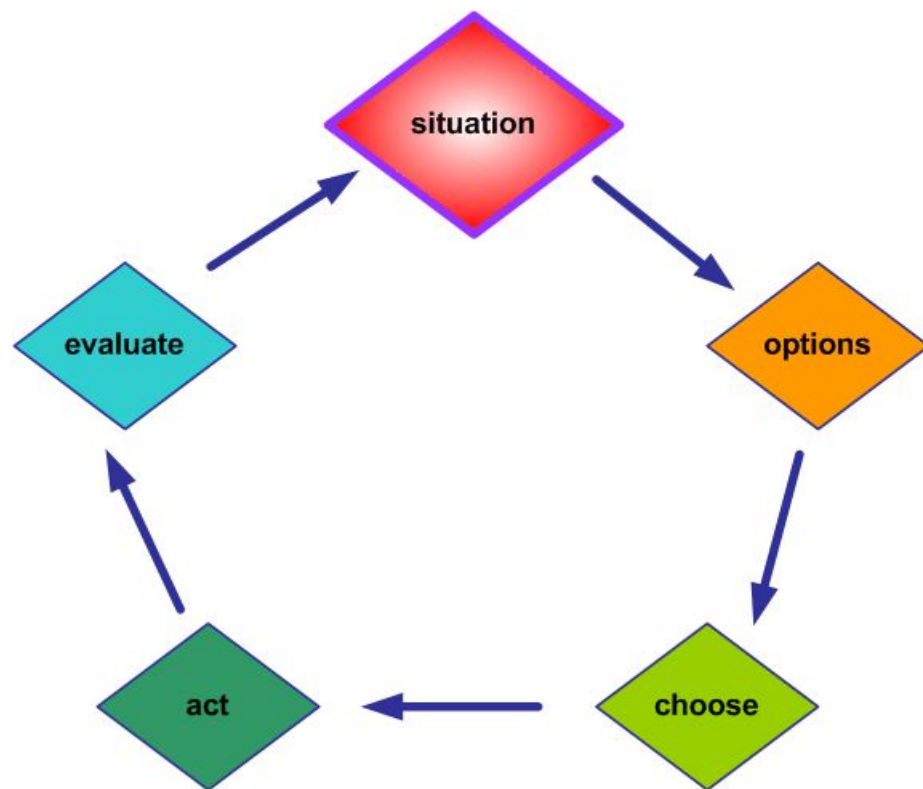
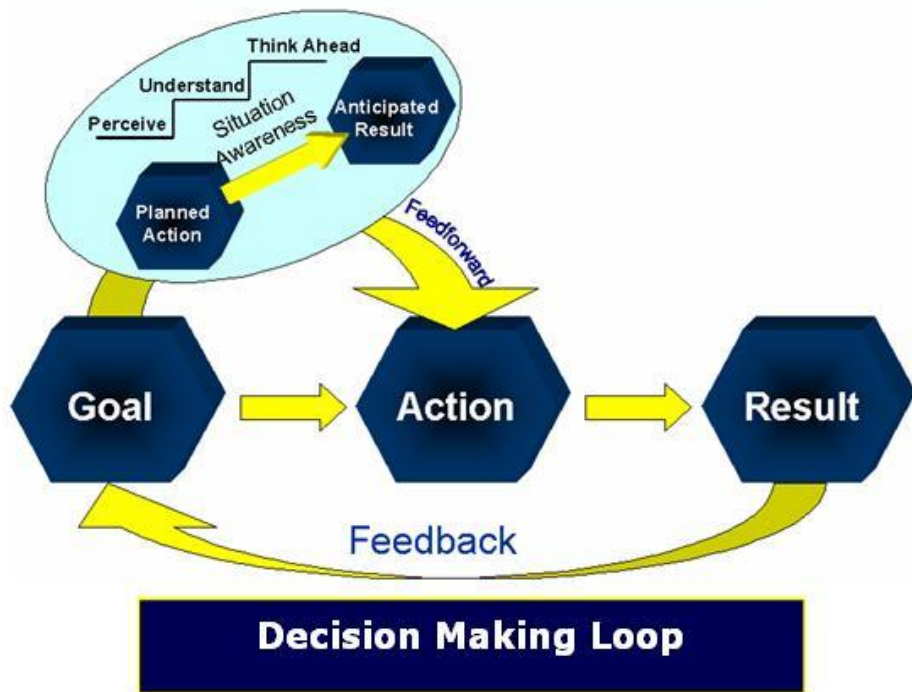
Identify actions to control the change.

Do (perform) the action.

Evaluate the success of the action.



Situational Awareness Process



3 P's of ADM



- **Perceive** the given set of circumstances for your flight. That means gathering all relevant information pertaining to your flight. This allows you to perceive the mission and the environment in which it will be flown. This step begins during preflight but continues throughout the flight.
- **Process** the information you gather, evaluate its impact on flight safety, and determine your best course of action. This analysis continues during the flight as you receive new information.
- **Perform** by implementing the best course of action. Performance results become information to be perceived and analyzed. Based on those results, pilots will decide whether to continue with the action or make a change.



Flight Risk Assessment Tools (FRAT)

Because every flight has some level of risk, it is critical that pilots are able to differentiate, in advance, between a low-risk flight and a high-risk flight, and then establish a review process and develop risk mitigation strategies. A FRAT enables proactive hazard identification, is easy to use, and can visually depict risk. It is an invaluable tool in helping pilots enhance their ADM skills and should be a part of every flight.

a FRAT tool that follows the PAVE checklist, covering questions on the Pilot, Aircraft, enVironment, and External Pressures.



Good Aeronautical Decision Making Skills Can Prevent Accidents



#FlySafe



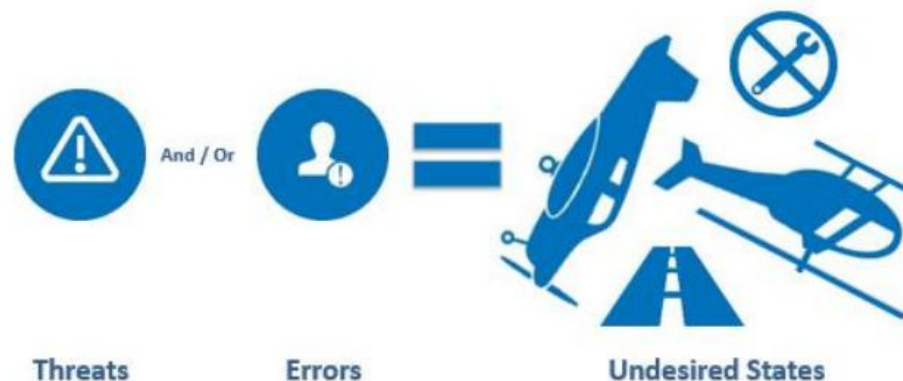
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Undesired Aircraft State

Threats and errors that are not detected and managed correctly can lead to an undesired aircraft state, which could be a deviation from flight path or aircraft configuration that reduces normal safety margins.

An undesired aircraft state *can still be recovered to normal flight* but if not managed appropriately, may lead to an outcome such as an accident or incident.

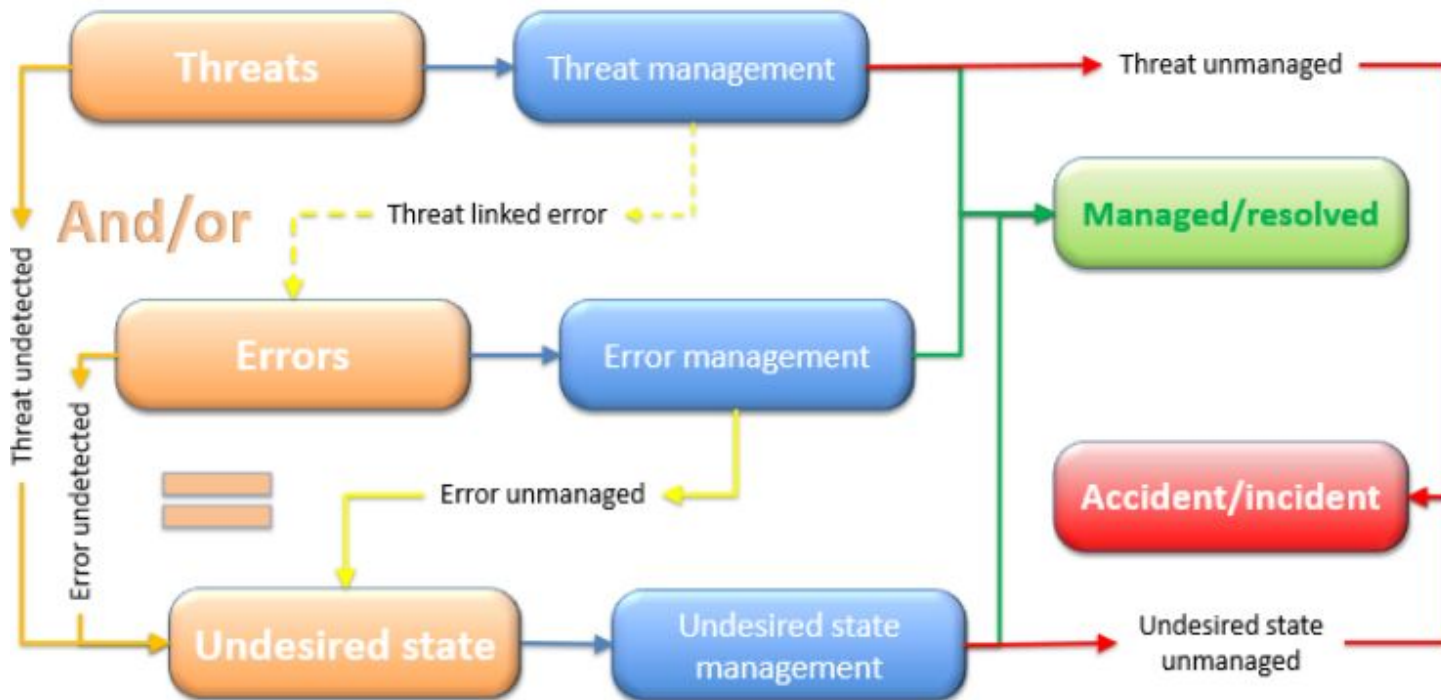
Safe flight in an aircraft requires recognition and recovery from undesired aircraft state in a very short time frame before an outcome such as loss of control, failure to achieve optimum performance or uncontrolled flight into terrain occurs.



Threat and Error Management Application

Using TEM, individuals can plan and apply appropriate countermeasures to identify and then manage threats and errors, to prevent them leading to an undesired state compromising the safety margins. Therefore, responding quickly and applying the appropriate actions to manage and/or resolve the threat and/or error ensures safety is maintained.





Assessing Risk

Risk is never “Zero” Therefore it must be assessed, then managed

Pilot-Self Assessment

- IMSAFE Checklist
- PAVE Checklist

Aircraft Preflight

- 360 Inspection

Flight Planning

- Pre-flight Briefing
- En-route
- Fuel Management
- Weather Briefing

✓	I'M SAFE CHECKLIST
	I llness—Do I have any symptoms?
	M edication—Have I been taking prescription or over-the-counter drugs?
	S tress—Am I under psychological pressure from the job? Worried about financial matters, health problems, or family discord?
	A lcohol—Have I been drinking within 8 hours? Within 24 hours?
	F atigue—Am I tired and not adequately rested?
	E motion—Am I emotionally upset?



Pilot



Aircraft



Environment



External Pressures



Pre-flight:

Just as you perform a number of tasks on a regular basis in preparation for flight (e.g. interpreting NOTAMs and MET information, checking fuel contents), pilots must include TEM as part of routine pre-flight planning and preparation.

A few minutes (or more) spent on the ground anticipating possible threats and errors associated with each flight will provide the opportunity to plan and develop counter measures. A good starting point is to ask what actions, conditions or event are likely to promote error, leading to the identification of internal and/or external threats applicable to that flight. This can reduce your workload airborne as you may have partially prepared yourself with how to deal with those threats and errors.



In flight:

When confronted by threats and/or errors a priority is to ensure the aircraft is in an appropriate configuration to optimize your ability to maintain control of the aircraft And flight path.

Monitor the progress of every sequence and abort if necessary

Do not fixate on threat or error management to the detriment of aircraft control

Identify and manage any undesired aircraft state and recover to planned flight



Post-flight:

Take a few minutes at the end of each flight to reconsider what threats, errors and/or undesired aircraft states were encountered during the flight. Ask yourself how well they were managed and what you will do differently to improve management of those threats and errors.

Record your threats, errors and/or undesired aircraft states and discuss them with more experienced pilots to assist with the development of improved TEM strategies.



Countermeasures

Threats and errors occur during every flight. Data from airline operations suggests that nearly half of fight-crew errors are not detected or responded to. It is very likely that this will be no better for charter operations.

TEM involves anticipating and calling out potential threats and errors as well as planning countermeasures in the self-briefing process at each stage of flight to prevent threats and errors becoming an undesired aircraft state. This needs to be done in a structured and simple way, without becoming complacent about commonly encountered threats such as weather, traffic, and terrain.



Countermeasures

There are three kinds of countermeasures:

- **Planning**

countermeasures including fight planning, briefing, and contingency planning.

- **Execution**

countermeasures including monitoring, cross-checking, workload and systems management.

- **Review**

countermeasures including evaluating and modifying plans as the fight proceeds, and enquiry and assertiveness to identify and address issues in a timely way.

Once you recognize an undesired aircraft state, you must use the correct countermeasure rather than fixate on the error.

TEM is therefore a continuous process; the practice of ongoing planning, appreciating, and controlling the progress of events to ensure a safe outcome.



ALWAYS REMEMBER !!

Threats come at you, Errors come from you

We are the Problem and the Solution

We often fail to recognize our own errors, overlook the errors of others and downplay Threats

Mitigate Risks by adopting Strategies and Countermeasures through Training and Practicing TEM

